Matching Markets Review 2

INFO 4220 – Networks II: Market Design

Format of this review

- I will go quickly over the key concepts
  - If anything of what I say today sounds unfamiliar, you should review the relevant lecture
- I have a contest with many PollEV questions at the end. Speed and accuracy matters for your final score (but it is just a game to test your understanding).

What We Have Studied

- This course is about who gets what and why
- Traditional Market Equilibrium
- One-to-one matching (musician-singer)
- Many-to-one matching (medical match)
- One-to-one assignment problems (dorm rooms)
  - Public, private, or mixed endowments
- Probabilistic assignments
- School choice problem
- Course allocation

Matching Markets

- We started from the supply and demand model in perfectly competitive markets
  - In thick markets prices adjust in response to supply and demand
  - Prices evolve to an equilibrium in which markets clear (quantity supplied at a give price = quantity demanded at a given price)
  - Prices act as coordination mechanism
- We then moved to problem where there are no prices: What if we want to match people to each other (musicians trying to form a duo)

Matching Markets

- First problem we looked at were singers trying to choose a musician to form a duo (and vice versa)
- There is no longer supply or demand:
  - We introduce preference orderings: Each person can create an ordered list of their most preferred matches
- There are no prices to coordinate exchanges or quantities sold:
  - We introduce a matching function
- There is no market equilibrium:
  - We introduce concept of stability

One-to-one matching

- How do we define stability:
  - Individual rationality: No one is matched to someone they find unacceptable
  - No blocking pairs: You cannot find two agents that would rather be matched to each rather than with whoever you matched them with
- Why do we want stability?
  - As otherwise agents will not accept the matchings or will choose not participate
- How do we find stable matchings:
  - Deferred acceptance algorithm

One-to-one matching

- How will singers and musicians behave when using the deferred acceptance algorithm
  - Strategyproof for the side proposing: It is a dominant strategy to reveal their true preferences
  - Optimality: When singers propose, matching is the singers-optimal matching (the most preferred stable matching for singers). When musicians propose, matching is the musician-optimal matching
  - Impossible to have strategy proofness for both sides and stability

Many-to-one matching

- What if one side (hospitals) can be matched to multiple agents from the other side (residents)
- We looked at how traditionally hospitals made offers to med students to become residents
  - And why it wasn't working
- We saw how we could adapt the deferred acceptance algorithm for these problems
  - We had to introduce the concept of responsive preferences
  - Now stability will require an additional requirement: Non-wastefulness

Preferences over sets and responsive preferences

- Hospitals are hiring several doctors. They may have preferences over groups of doctors. We introduced preferences over sets $P_h^{\#}$
  - $\{d_1, d_2\} P_h^{\#} \{d_3, d_4\}$ Means that hospital h prefers hiring doctors 1 and 2 over doctors 3 and 4

- Responsive preferences:
  - The hospital should compare: {A, X, Y} and {B, X, Y}
  - The responsive preferences hypothesis implies that it is sufficient to compare Dr. A and Dr. B:

$$\{A, X, Y\} P_h^{\#} \{B, X, Y\} \Leftrightarrow A \; P_h \; B$$

Stability in many-to-one assignments

- Individual rationality:
  - No doctor would prefer to be unmatched rather than their current matching
  - No hospital is matched to a doctor they consider unacceptable
- No blocking pairs:
  - You cannot find a pair (d,h) such that d is not matched to h, d would prefer to be matched to h than to its current match, and hospital h would like to change one of the doctors in their matching for d
- Non-Wastefulness:
  - If a doctor likes hospital h better than their match (and the hospital does not find them unacceptable), then hospital h must be full

Agent Behavior in Many-to-One Matchings

- Strategyproof for doctors when they propose
- Not strategyproof for hospitals, even when they are proposing
- When doctors propose, D.A. finds the doctors-optimal match. When hospital propose, D.A. finds hospital-optimal match
- We also discussed some special applications: What happens if you use an algorithm that does not create stable matchings, what happens with couples, and the rural hospital problem

Assignment Problems

- In matching problems both sides have preferences
- How about problems where one side is an object
  - Objects don't have preferences. We don't need to care if the object likes who they are assigned to
  - We call this assignment problems
- Although objects don't have preferences, they could have owners: We call this endowment
  - Public endowment: Objects belong to everyone
  - Private endowment: Objects belong to individuals
  - Mixed endowment: Some objects are private, some public

Evaluating Assignments

- Individuals have preferences over objects
- Objects don't have preferences
- Who look for assignments that are Pareto Efficient:
  - An assignment is Pareto Efficient if there is no other assignment where non is worse off and at least someone is better off.
- In some problems we may will to consider that objects may have an owner, or that some people have priority to get an object

Matching vs Assignment Problems

- In matching problems, we care about the preferences of both side.
  - We look for matchings that are stable: Can we find people that would like to be matched to each other (or be unassigned) rather than being with their current match
    - Individually rational, No-blocking pairs, Non-wasteful (in many-to-one problems)
- In assignment problems, we care about only one side
  - We look for assignments that are efficient: Can we find an assignment that would make someone better off without making anyone worse off?

Serial Dictatorship

- Simple mechanism and when used in problems without endowments is:
  - Efficient
  - Strategyproof
  - Individually rational
- Can also be used in problems **with endowments**:
  - Still strategyproof
  - Still efficient
  - But someone could end up with an object they like less than their initial endowment. Thus, **not individually rational**

Individual Rationality and The Core

A	B	C	D	E	< Endowment
Boxing	Chorus	Finance	Gaming	Movie	
C	A	B	E	**E**	
B	E	C	**A**	F	
A	**C**	E	D	B	
F	B	**F**	C	D	
D	D	D	F	C	
E	F	A	B	A	

- What if we use a serial dictatorship in the order: Movie, Gaming, Chorus, Boxing, Finance
- There are 2 problems with this assignment:
  - Finance got something worst than they initially had: Not individually rational, they don't want to participate
  - Boxing, Chorus, and Finance could trade between them and improve their initial endowment: Not in the core, they will prefer to trade between them rather than participating in the market

Top Trading Cycle

- When there are endowments, the top trading cycle algorithm works better
- Algorithm that allow agents to trade objects:
  - Efficient
  - Strategyproof
  - Has the advantage over serial dictatorship of also being individually rational

Theorem: An assignment mechanism is strategyproof, efficient, and individually rational iff it uses the top trading cycle algorithm

Mixed Public-Private Endowments

Typical example: Dorm room allocation

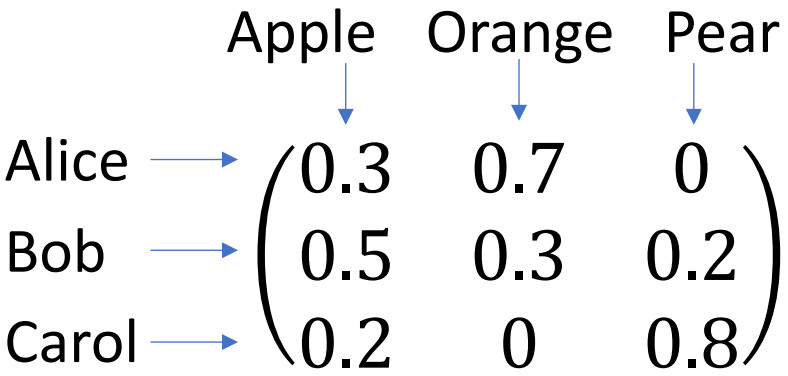
- Random serial dictatorship with squatting rights: Not individually rational nor efficient
- Random serial dictatorship with waiting lists: Not efficient
- MIT-NH4: Still not efficient
- Mechanisms that are strategy proof and finds efficient and individually rational assignments
  - TTC w/ mixed endowments and YRMH-IGYT

Probabilistic Assignments

- We have been assuming that assignments are deterministic, but they don't need to be
- For example, we may need some randomness to break ties
- Or we may want to introduce randomness to increase "fairness"
  - For example, randomize order in serial dictatorship
- We need:
  - A way to express/represent random assignments
  - A way to evaluate random assignments
  - Algorithms to find random assignments

How to represent random assingments

Assignments are now probabilities of getting an object



Alice gets:

- An apple with probability 0.3: $\beta_{Alice,Apple} = 0.3$
- An orange with probability 0.7: $\beta_{Alice,Orange} = 0.7$
- A pear with probability 0: $\beta_{Alice,Pear} = 0$

Representing/evaluating random assignments

The random assignment:

$$\begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/2 & 0 & 1/2 \\ 0 & 3/4 & 1/4 \end{pmatrix}$$

Is equivalent to the following deterministic assignments with the given probabilities:

$$\frac{1}{2} * \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{4} * \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{4} * \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Representing/evaluating random assignments

- We need a way to compare how an individual does under two different random assignments:
  - We introduced the concept of stochastic dominance
- We also need a way to evaluate the efficiency of a mechanism. This can be done:
  - Ex-ante: Before the random draw happens (stochastic dominance)
  - Ex-post: After the random draw happens (compare actual assignments that would result)

Ex-Ante vs. Ex-Post

- Remember the game we did
- I asked for 2 volunteers
- I said I was going to flip a coin and whoever guessed heads or tails gets a bonus for PS
  - Ex-ante: Compare the odds of getting the bonus
  - Ex-post: We flipped the coin. There were 2 possible scenarios
- Then we re-did the game changing the rules. Winner of round 1 had to guess 2 times in a row to get the bonus

Algorithms for Probabilistic Assignments

- Random serial dictatorship
- TTC with random endowments
- To produce a random assignments with these algorithms above you need to consider all possible random assignments and then sum the probabilities of each possible assignment
- The two above actually produce the same random assignments. And both do not guarantee ex-ante efficiency
- Solution to find ex-ante efficient assignments: Use eating algorithms (which is a family of algorithms)

School Choice

- Similar to a many-to-one assignment, but needs some refinement
  - Schools are public goods, they cannot discriminate (everyone is acceptable)
  - Schools don't have preferences, but some schools can have priorities
- Stability will now require one new condition:
  - Individual rationality
  - Non-wastefulness
  - No justified envy
- We evaluate efficiency using Pareto definition and considering only students' preferences

Stability in School Choice

- It is individually rational: For each student $i \in I, \mu(i)$ is weakly preferred to the option of being unassigned
- It is non wasteful: For each student $i \in I$:
  - $sP_i\mu(i) \Rightarrow |\mu(s)| = q_s$
- The is no justified envy: There will be justified envy if a student prefers school s to their assignment, and there is another student assigned to school s that has a ***strictly lower*** priority

Blocking Pairs vs. Justified Envy

- In matching problems, we wanted assignment that did not have a blocking pair
  - We did not want a situation in which we had 2 agents that would rather be matched to each other than with whoever you matched them with
- In school choice, schools do not have preferences. But there are priorities. So, what we want to avoid is justified envy
  - We don't want a situation in which you have a student s_1 that envies the school student s_2 got assigned to, and that s_1 has a higher priority than s_2 at that school
- Blocking pairs and justified envy are similar, but conceptually different

Efficiency in School Choice

An assignment μ is efficient if there is no other assignment μ' such that:

- All students weakly prefer μ' to μ
  - All students are either indifferent between μ' and μ or prefer μ' to μ
- There is at least one student who strictly prefers μ' to μ
  - At least one student is not assigned to the same school in μ and μ', and she prefers the school she is assigned in μ'

Stability vs Efficiency

- In the school choice problem stability and efficiency are not compatible (note that this doesn't mean both things can't happen at the same time)
- There is no strategy-proof mechanism that selects the efficient and stable matching whenever it exist
- If you want to get stability: Use deferred acceptance
- If you want to **get efficiency: Use top trading cycles**

How do School Districts Decide?

- We discussed how School Choice was reformed in Boston and NYC
- They both ended up using deferred acceptance, but for different reasons
- Boston had a strategy proofness issue (they were using immediate acceptance)
- NYC had a congestion issue (the had a decentralized system)
- Practice in school choice has been to move from very bad mechanisms, to better (but not perfect) mechanisms
- The decision of which mechanisms to use is social/political

The Problem of Weak Priorities

- Schools' priorities are weak. We cannot really use DA and TTC with weak priorities, so we break ties randomizing
- This introduces an additional source of inefficiencies
  - But that can be solved with improvement cycles
- We could break ties with single or multiple tie breaking
  - Single tie breaking: The same lottery is used to break ties at all schools
  - Multiple tie breaking: Each school runs their lottery to break ties
  - While multiple tie breaking "seems" fairer, single works better in practice (more students get their most preferred schools)

Course Allocation

- This is a many-to-many assignment problem. A lot more complicated than problems we saw before:
  - No obvious choice of algorithms. Before we could have an algorithm that was either: strategyproof, efficient, or stable
  - Combinatorial assignment problem: Many possible assignments are possible
  - Moreover, some courses are complements, while other are substitutes: We cannot really use responsive preferences
- To analyze assignments in this context we needed a new definition: Non-bossy assignment

Serial Dictatorship in Course Assignment

Non-bossy means that nobody can change other individuals' assignment without changing their own assignment

Theorem (Pápai):

A many-to-many assignment mechanism is strategyproof, non-bossy, and pareto efficient if, and only if, it is a serial dictatorship mechanism

Course Allocation in Practice

- Many universities use a bidding system:
  - Students bid for courses
  - Run a serial dictatorship in the order of bids (from higher to lower, if two are equal determine randomly)
  - In their turn student can only choose the course the bid for (if it is full, they lose their turn)
- Not strategy-proof, and bids do not reflect the true "prices" of courses
  - Students may bid low for courses they like and expect will not fill up
  - And bid high for courses they don't like as much, but they expect will fill up

Course Allocation in Practice

- A better solution than bidding is using deferred acceptance with bidding (Gale-Shapley Pareto Dominant Market Mechanism):
  - Combines deferred acceptance with bids
  - Students submit list of preferences and bids for courses
  - Run deferred acceptance using the preferences. If a course needs to reject students, it rejects the lowest bids (if two are equal, it is determined randomly)
- Submitting true preferences is a dominant strategy
- How much to bid is a strategic decision

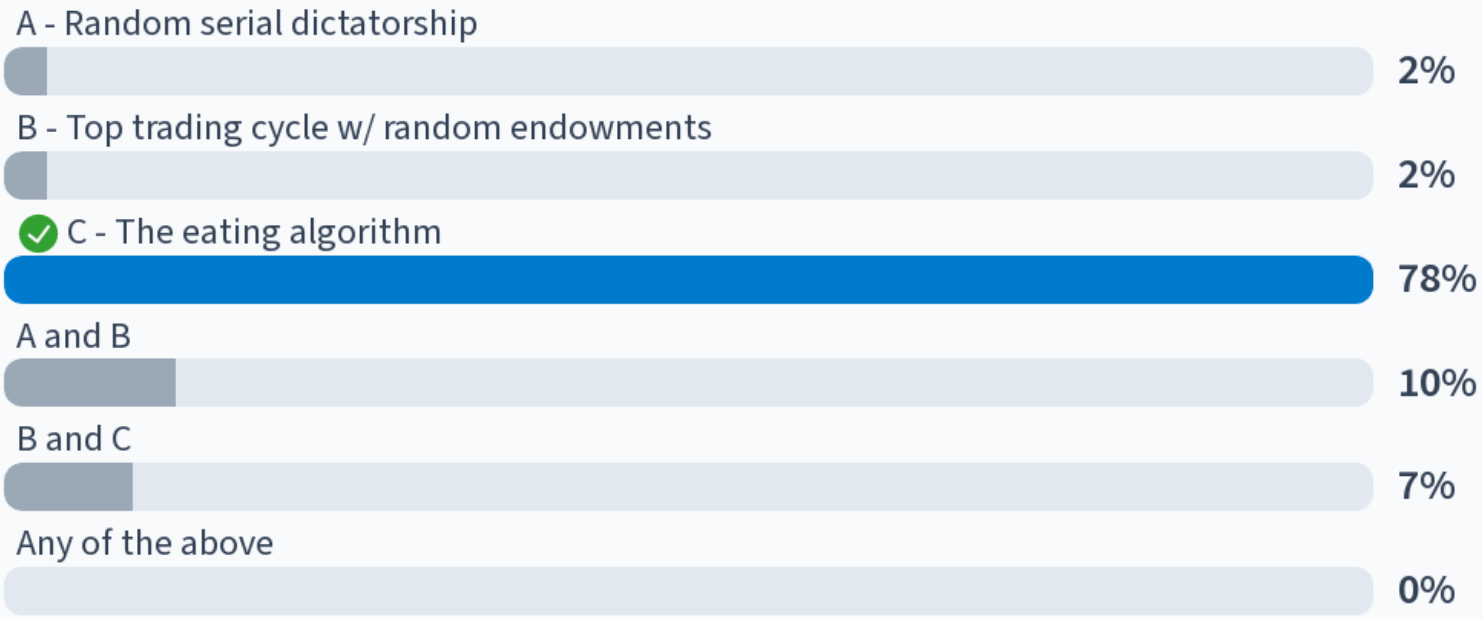
Course Allocation in Practice

- HBS draft mechanism:
  - Serial dictatorship choosing one course at a time
  - Order of dictators is inverted at each round
- Students engage in strategic behavior
- It may not be strategyproof or efficient, but in practice works better (students like the outcome better on average) than a random serial dictatorship (with student picking full schedules during their turn)

Course Allocation in Practice

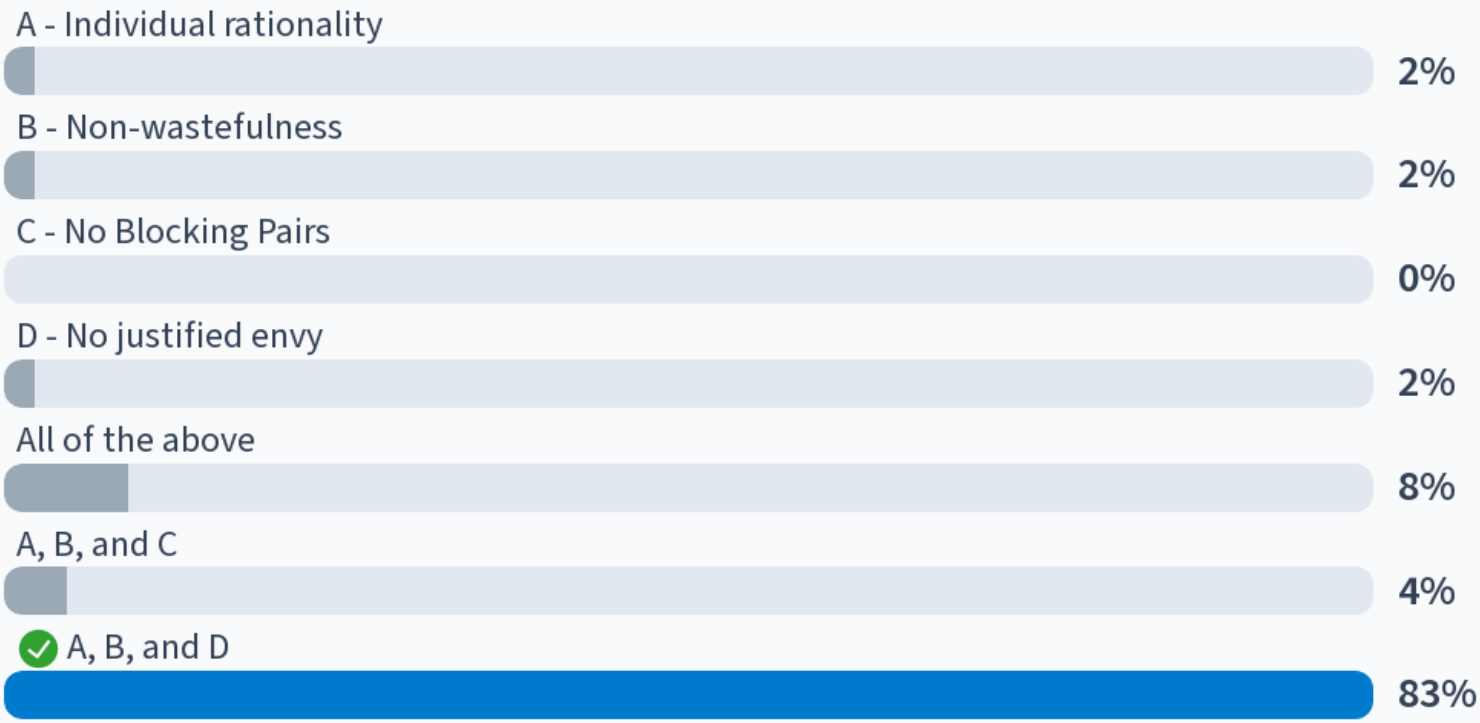
- The Wharton School uses the “Approximate Competitive Equilibrium with Equal Incomes”
- Students submit bids over “schedules”
  - Not really, as considering all schedules would be impossible
  - Students submit bids, and adjustment for combinations of courses. These are used to compute bids over schedules
- They only look for an “approximate” equilibrium
- Outcomes produced by this algorithm are preferred by students (compared to traditional bidding mechanism)

In a probabilistic assignment problem, which of these algorithms produce ex-ante efficient assignments



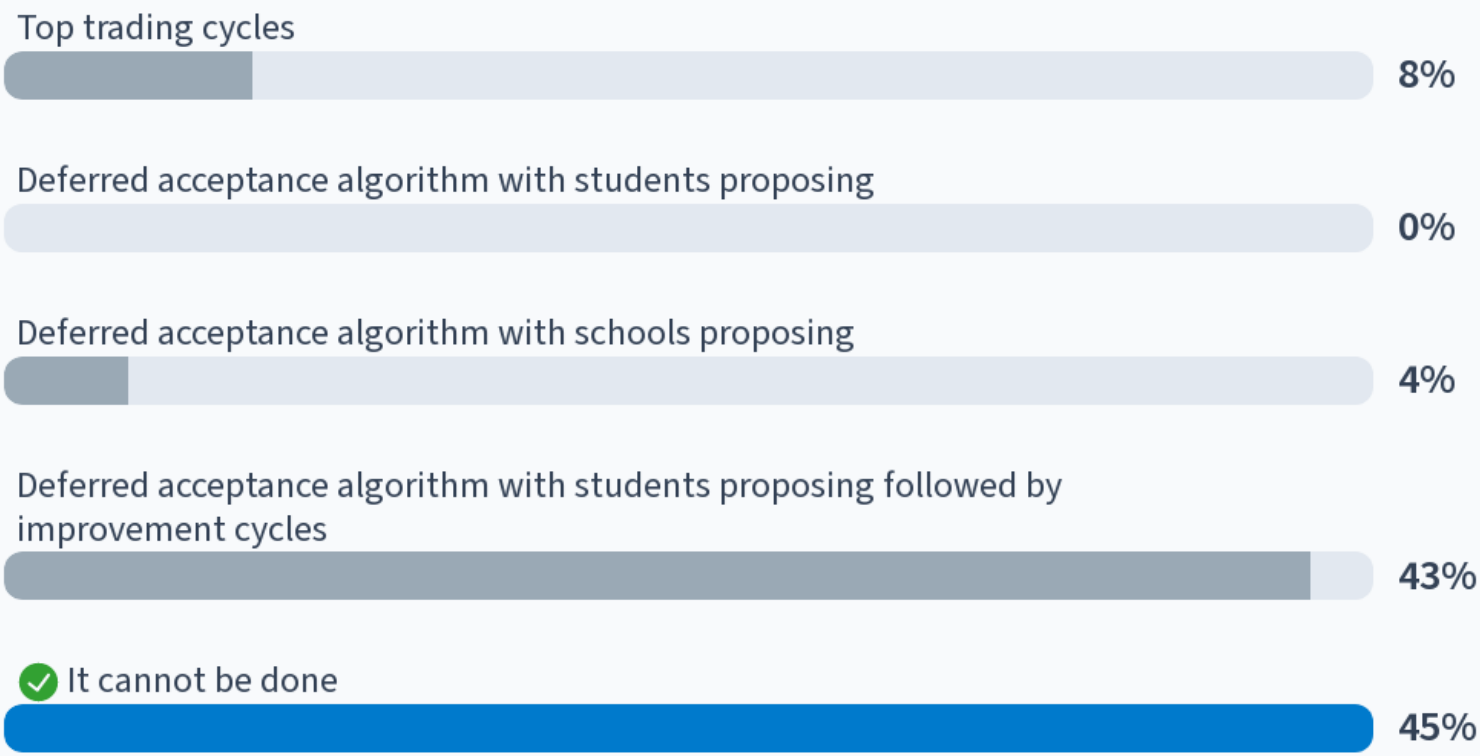
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

In the school choice problem, stability requires:



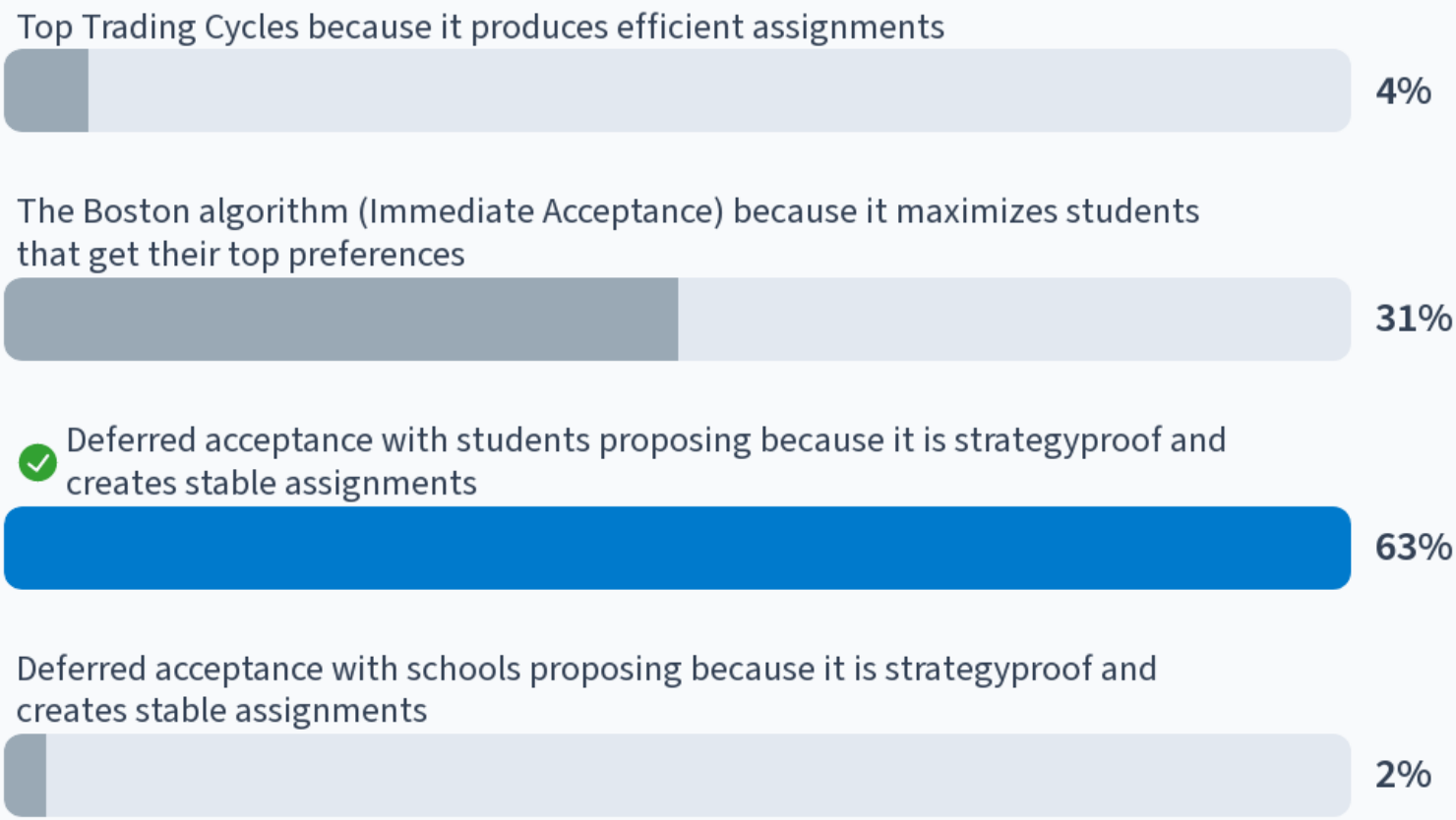
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

The school district of Gotham City is asking you which algorithm should they use to assign students to schools. They want an algorithm that always creates assignments that are efficient and stable, which algorithm would you recommend?(priorities are weak)



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

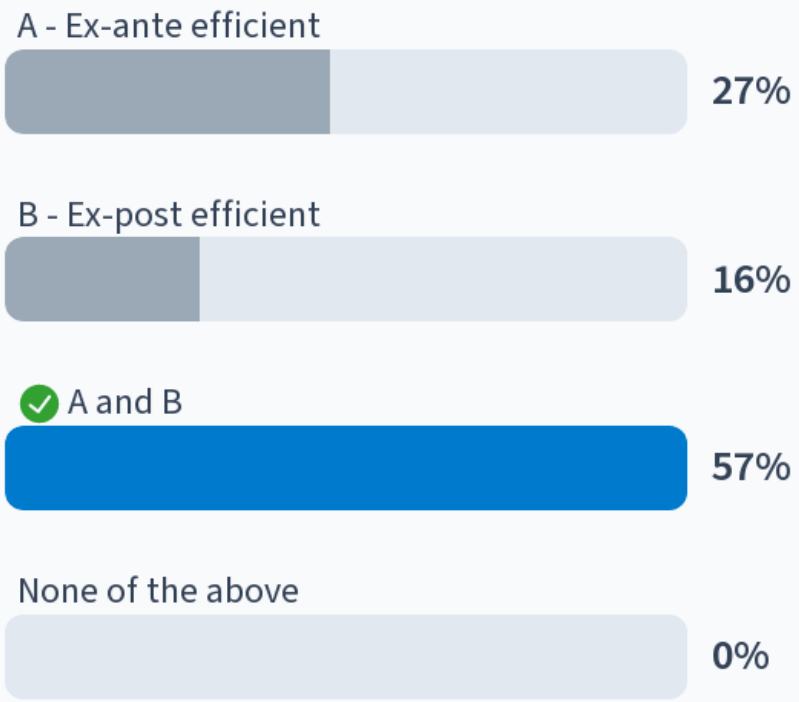
Which algorithm did the Boston school district choose and why



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

The preferences of Alice and Bob over Apples and Oranges are as in the table. A probabilistic assignment that assigns 50% probability to Alice getting the Apple and Bob the Orange, and 50% to Alice getting the Orange and Bob the Apple is:

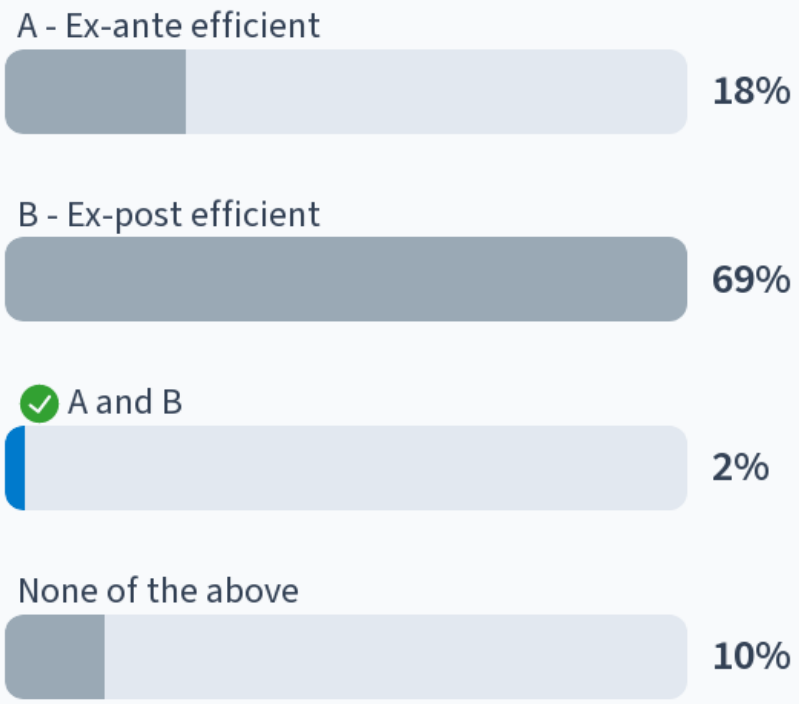
Alice	Bob
Apple	Apple
Orange	Orange



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

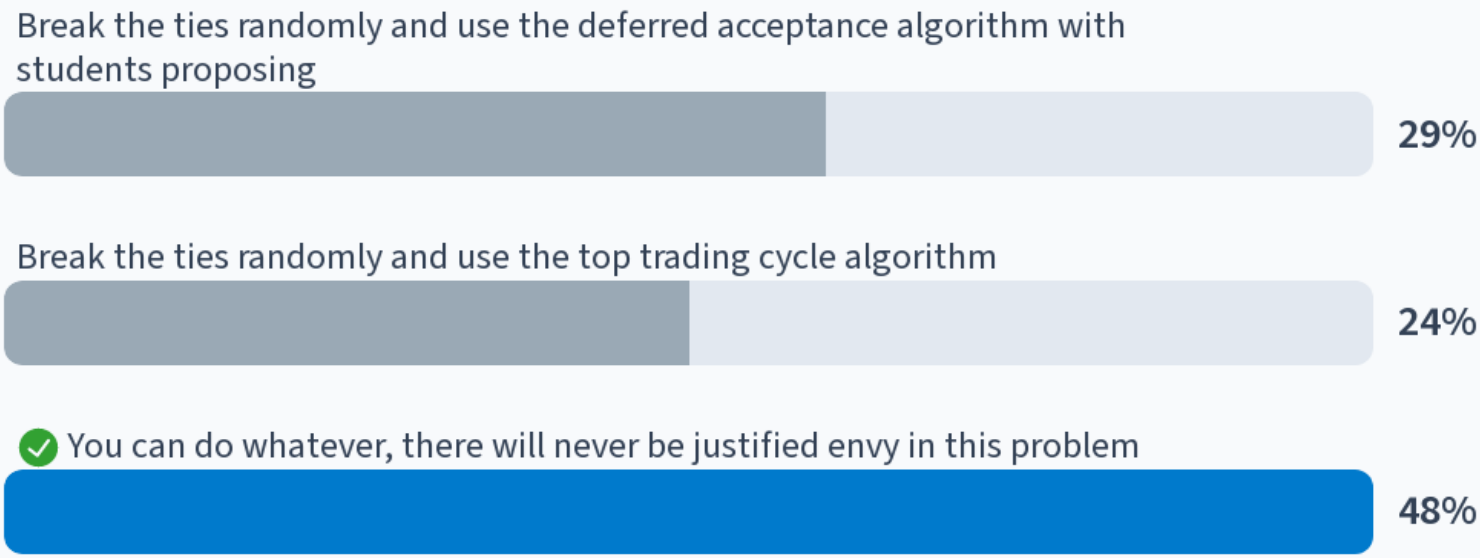
The preferences of Alice and Bob over Apples and Oranges are as in the table. A probabilistic assignment that assigns 100% probability to Alice getting the Apple and Bob the Orange, and 0% to Alice getting the Orange and Bob the Apple is:

Alice	Bob
Apple	Apple
Orange	Orange



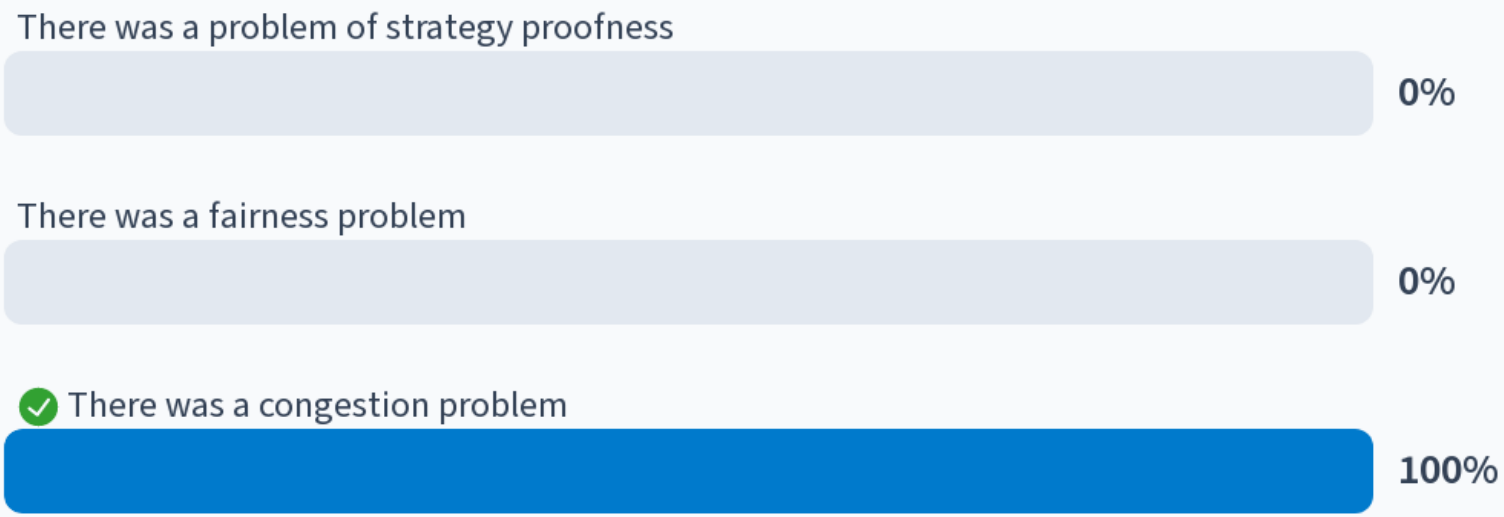
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

In school assignment problem there 3 schools (capacity of 2 each) and 6 students. All the students have the same priority (weak) at each of the schools. How can you ensure the will be no justified envy?



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What was the problem that New York City was trying to solve in their redesign of their school assignment problem



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Is there are problem with the following assignment?

Endowments →	A	B	C	D	E
	P_{Alice}	P_{Bob}	P_{Carol}	P_{David}	P_{Ellie}
	B	C	A	**C**	**A**
	E	A	**E**	A	C
	D	**D**	D	E	B
	C	E	B	B	D
	A	B	C	D	E

A - No, there is nothing wrong with it

0%

B - Yes, it is not individually rartional

0%

C - Yes, it is not in the core

0%

B and C

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**